

NON-PARAMETRIC EMPIRICAL BAYES ESTIMATION OF QUANTILES

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Summary : A new form of Bayesian estimate for the quantiles of a distribution drawn from a Dirichlet process prior is derived. This estimator is applied to the empirical Bayes situation, in which the prior measure parameterizing the Dirichlet process is itself estimated on the basis of samples from different distributions, all from the same Dirichlet process prior. The performance of this non-parametric empirical Bayes (NPEB) estimator in estimating quantiles from relatively small sample sizes, is compared, by means of a Monte Carlo study, to that of a number of other estimators. It is found that the NPEB procedure is very nearly as efficient as normal theory estimators when the assumption of normality in fact applies, while showing substantial gains over the other methods when the distributions are contaminated. Finally, an application of the NPEB procedure to streamflow data from dams in the Transvaal is discussed.

1. Introduction

It is at times required to estimate tail quantiles of a distribution from relatively very sparse data. For example, hydrologists are frequently concerned with such quantities as the 20-year and 50-year flood, i.e. the $1/20$ and $1/50$ quantiles of the distribution of annual maximum river levels. Frequently, however, measurements may only be available over little more

than 20-50 years! In the light of such sparsity of data, it is common practice to assume a parametric form for the distribution of the random quantity, and to estimate the parameters using the full data set available. Thereafter, the desired quantiles can be estimated using the fitted distribution. Some refinement of this procedure, aimed at obtaining best estimates of the quantiles themselves (rather than obtaining best estimates of the parameters, and naively using the quantiles of the fitted distribution), have been proposed for the normal distribution (see for example, Dyer, Keating and Hensley (1977)). The estimates remain, nevertheless, very sensitive to the assumption of distributional form.

There is an intuitive feeling, at least, that data from other similar situations should be able to improve the quality of quantile estimates. In the above example, data from other rivers having similar topography, climatic conditions, etc., might be expected to identify at least whether the maximum in the observed 50 years, for example, is atypically high or low relative to the mean - and such information should improve quantile estimates. Within the context of parametric models, empirical Bayes procedures have been used to improve parameter estimates in this way (cf. Kuczera (1982)). More generally, however, such related data may contain information concerning the distributional form itself, especially in the tails. This would suggest a non-parametric empirical Bayes procedure, and it is this possibility which we explore in the present paper.

As a basis for the non-parametric empirical Bayesian analysis, we shall use the Dirichlet process concept due to Ferguson (1973, 1974). Although this approach needs to be used with some caution (see, for example, the discussion of Diaconis and Freedman (1986)), it appears nevertheless to remain the most practically convenient framework for non-parametric Bayesian analysis. It is convenient to review here briefly, the key concepts of the Dirichlet process, which will also serve to introduce some notation.

Let X be a real-valued random variable, and let $F(\cdot)$ denote the distribution function of X . Consider any arbitrary partitioning of $\mathbb{R}$ into

disjoint subsets $I_1, I_2, \dots, I_K$, such that $I_1 \cup I_2 \cup \dots \cup I_K = \mathbb{R}$. For $k=1, 2, \dots, K$, define:

$$p_k = P[X \in I_k].$$

The basic assumption underlying the Dirichlet process approach is that there exists a finite measure $\alpha(\cdot)$, defined on appropriate subsets of the real line (e.g. the Borel sets), such that for any partitioning of $\mathbb{R}$ as above into measurable subsets, the prior distribution of $(p_1, p_2, \dots, p_K)$ is Dirichlet with parameters $\alpha_k = \alpha(I_k)$; i.e. the joint prior p.d.f. of $(p_1, p_2, \dots, p_K)$ is given by

$$\pi(p_1, \dots, p_K) = \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_1)\Gamma(\alpha_2)\dots\Gamma(\alpha_K)} \prod_{k=1}^K p_k^{\alpha_k-1} \quad \left(\sum_{k=1}^K p_k = 1; p_k \geq 0 \right)$$

$$= 0 \quad \text{(elsewhere)}$$

where

$$\alpha_0 = \sum_{k=1}^K \alpha_k = \alpha(\mathbb{R}) < \infty, \quad \text{by definition.}$$

Note that in the special case in which $\mathbb{R}$ is partitioned into two subsets, I and $\bar{I}$ say, the prior distribution on $q = P[X \in I]$ is Beta with parameters $\alpha(I)$ and $\alpha_0 - \alpha(I)$.

The probability distribution on $F(\cdot)$ induced by these Dirichlet priors for any arbitrary partitioning is termed a Dirichlet process prior. It is easily seen (Ferguson 1973, 1974) that the posterior distribution for $F(\cdot)$, conditional on $X_1 = x_1, \dots, X_m = x_m$, is also a Dirichlet process, with associated measure given by:

$$\bar{\alpha}(I) = \alpha(I) + \sum_{j=1}^m \delta_{x_j}(I)$$

where:

$$\delta_x(I) = 1 \quad \text{if } x \in I$$

$$= 0 \quad \text{otherwise.}$$

Note that $\bar{\alpha}_0 = \bar{\alpha}(\mathbb{R}) = \alpha_0 + m$.

The Dirichlet process prior and posterior can form the basis of a non-parametric Bayesian Analysis (Ferguson, 1973). In section 2 we discuss the application of these results to estimation of quantiles under a minimum squared error criterion, for any given Dirichlet process measure $\alpha(\cdot)$. The resulting estimator is easy to implement when $\alpha(\cdot)$ places all mass on a finite set of points, which is precisely the situation arising in *empirical Bayes* estimation. Within the context of the Dirichlet process, an empirical Bayes approach requires that the prior measure $\alpha(\cdot)$ be estimated from empirical data. Formally, suppose we have samples size m from each of n populations, i.e. $X_{ij} = x_{ij}$ for $i=1, \dots, n$; $j=1, \dots, m$, where for each i , X_{ij} has the distribution function $F_i(\cdot)$, say. We suppose that the populations are sufficiently similar that the prior distribution for each $F_i(\cdot)$ can be represented by a Dirichlet process with common associated measure $\alpha(\cdot)$. In this case, the full set of data can be used to estimate a suitable $\alpha(\cdot)$ in some sense, which may then be used for inferences concerning each individual $F_i(\cdot)$.

Details of such an empirical Bayes approach are given in section 3. We note here, however, that the same approach has been used for example by Tiwari and Zalkikar (1985). The problem we are investigating differs from theirs in two important aspects, viz.:

- (1) They consider inferences concerning parameters which are functionals of $F(\cdot)$, i.e. of the form

$$\phi = \int_{-\infty}^{\infty} h(x) dF(x) ;$$

quantiles are not expressible in this form in general.

- (2) They examine asymptotic properties of estimators as $n \rightarrow \infty$; in the motivating examples above, which are our primary interest, the number of related cases, i.e. n , is likely to be small.

In fact, for small numbers of populations, and for estimation of quantiles, it does not appear to be possible to obtain properties of estimators analytically. For this reason, a numerical study of the estimators proposed in section 3 is reported in section 4.

Finally, the procedures are applied to streamflow data from six dams in the Transvaal. In each case, data for the same 40 years were available; data from 20 years were used to obtain quantile estimates, and these were compared with actual extreme values in the remaining 20 years. Details of this comparison are given in section 5.

2. Estimation of quantiles under a minimum squared error criterion

Suppose that the distribution function $F(\cdot)$ is itself distributed according to a Dirichlet process prior, with measure $\alpha(\cdot)$. It is immaterial at this stage whether $\alpha(\cdot)$ is a prior or posterior measure. Define θ to be the upper β -th quantile of F , i.e.:

$$F(\theta) = 1 - \beta$$

for some given $\beta \in (0, 1)$.

Ferguson (1973, p. 225) discusses estimation of θ , for a loss function based on absolute deviations from the true value (possibly weighted differently for positive and negative deviations). This estimate is shown to be a quantile of the expectation of F . The required quantile is a complicated function of β , the weights, and $\alpha(\mathbb{R})$, although Ferguson has tabulated certain values.

In many estimation problems, however, loss proportional to squared deviations is more appropriate (since small deviations are relatively less

serious than large, especially, for example, in engineering design). It is well known that the relevant estimator is $E[\theta]$, where this expectation is taken with respect to the distribution of θ induced by the Dirichlet process.

Suppose now there exists some $\theta_L > -\infty$ such that $\alpha[(-\infty, \theta_L)] = 0$. In this case, it is well known that

$$(1) \quad E[\theta] = \theta_L + \int_{t=\theta_L}^{\infty} P[\theta > t] dt.$$

Furthermore:

$$P[\theta > t] = P[F(t) < 1 - \beta]$$

while $F(t)$ has the Beta distribution with parameters $\alpha_t = \alpha[(-\infty, t)]$ and $\alpha_0 - \alpha_t$, where $\alpha_0 = \alpha(\mathbb{R})$. Thus:

$$(2) \quad P[\theta > t] = \frac{\int_0^{1-\beta} z^{\alpha_t-1} (1-z)^{\alpha_0-\alpha_t-1} dz}{B(\alpha_t, \alpha_0 - \alpha_t)}$$

where $B(\alpha_t, \alpha_0 - \alpha_t)$ is the complete Beta function with parameters α_t and $\alpha_0 - \alpha_t$.

In general, evaluation of $E[\theta]$ requires non-trivial numerical evaluation. One special case is however easily evaluated: this is the case in which $\alpha(\cdot)$ concentrates all mass on a finite set of points $w_1, w_2, \dots, w_N$ say. This is precisely the situation which arises in all empirical Bayes work, and is thus of particular interest to us. In this case for any $1 \leq \ell < N$, and for all $t \in (w_\ell, w_{\ell+1}]$:

$$\begin{aligned} \alpha_t &= \alpha(\{w_1\}) + \alpha(\{w_2\}) + \dots + \alpha(\{w_\ell\}) \\ &= \alpha_\ell, \text{ say.} \end{aligned}$$

and $Var_{\alpha}[\cdot]$ will represent expectation and variance induced by the Dirichlet process parameterized by $\alpha(\cdot)$. The predictive distribution for a random variable X drawn from a distribution F generated from the above Dirichlet process is given by $E_{\alpha}[F(\cdot)]$, where

$$(4) \quad E_{\alpha}[F(x)] = \alpha((-\infty, x]) / \alpha_0.$$

Let $w_1 \leq w_2, \dots, \leq w_N$, where $N = nm$, be ranked values of the entire ensemble of x_{ij} values. Since each X_{ij} has predictive distribution function given by (4), it is clear that a consistent empirical estimate for $\alpha(\cdot)$, $\hat{\alpha}(\cdot)$ say, will for any fixed α_0 be given by:

$$(5) \quad \hat{\alpha}(\cdot) = \frac{\alpha_0}{N} \sum_{\ell=1}^N \delta_{w_\ell}(\cdot).$$

The problem arises in selecting α_0 . It has been noted by Zehnwirth (1981) that:

$$\alpha_0 = E_{\alpha}[Var_i(X_{ij})]/Var_{\alpha}[E_i(X_{ij})]$$

and thus that an obvious empirical "estimate" for α_0 is:

$$\begin{aligned} \hat{\alpha}_0 &= \frac{\sum_{i=1}^n \sum_{j=1}^m (X_{ij} - \bar{X}_{i.})^2 / n(m-1)}{\sum_{i=1}^n (\bar{X}_{i.} - \bar{X}_{..})^2 / (n-1)} \\ &= m/F \end{aligned}$$

where F is the F-ratio statistic in the one-way ANOVA, testing for differences between populations. Zehnwirth points out, however, that this estimate is biased, and therefore suggests using $m/(F-1)$ when $F > 1$. When $F \leq 1$, he suggests $\hat{\alpha}_0 = 0$ as an estimator. Apart from the discontinuity (as a function of F) implied, there is a more fundamental problem with this. The case of $F \leq 1$ is evidence in favour of no difference between populations, i.e. all F_i equal. This in turn implies that the Dirichlet process generating the F_i shows very little variation, which corresponds to large values of α_0 . For this reason we shall use the following empirical estimate for α_0 :

$$(6) \quad \begin{aligned} \hat{\alpha}_0 &= m/(F-1) \\ &= m/\epsilon \end{aligned} \quad \left. \begin{aligned} F &\geq 1 + \epsilon \\ F &< 1 + \epsilon \end{aligned} \right\}$$

for some arbitrary but small $\epsilon > 0$. From the results given by Zehnwirth (1981, p. 223), it is easily seen that $\hat{\alpha}_0$ is consistent (for large n , irrespective of m) provided that

$$\epsilon \leq m \text{Var}_{\alpha}[E_i(X_{ij})]/E_{\alpha}[\text{Var}_i(X_{ij})] = m/\alpha_0.$$

Consistency can thus be ensured, provided an upper bound $A \geq \alpha_0$ can be specified *a priori*, by setting $\epsilon = m/A$.

The empirical Bayes form of the Dirichlet process posterior for F_i thus allocates all mass on to the points $w_1, w_2, \dots, w_N$, and thus the required quantile of F_i can be estimated using (3). The relevant (estimated) Dirichlet posterior measure places a mass of $\hat{\alpha}_0/N$ at each observation point x_{kj} for $k \neq i$, and a mass of $1 + \hat{\alpha}_0/N$ at each point x_{ij} . The procedure may be summarized in the following:

Algorithm

- (1) Define $w_1, \dots, w_N$ as the order statistics of the full set of $N = nm$ observations.
- (2) Select an $\epsilon > 0$ based on some practical and plausible upper limit on α_0 , and hence estimate α_0 using (6).
- (3) Estimate the required quantile θ from (3), where G_ℓ is obtained from the right-hand side of (2) by substituting $\hat{\alpha}_0 + m$ for α_0 ; and

$\ell \hat{\alpha}_0 / N + (\text{number of observations from population } i \text{ not exceeding } w_0) \text{ for } \alpha_i.$

4. Numerical comparisons with other estimators

The behaviour of the above non-parametric empirical Bayes (NPEB) estimator for $\beta = 0,1$; $0,05$; and $0,02$ respectively, was investigated by means of a Monte Carlo study, in the cases of sampling from normal and from contaminated normal distributions. In particular, the mean squared error (MSE) of estimation of θ , was compared with that of three other estimators, as follows:

- (i) the best invariant estimate (BIE) assuming the data are normally distributed, as described in Dyer *et al.* (1977): when the data are in fact normal, this estimator has lowest MSE amongst sampling theory estimators, i.e. estimators using information in the relevant sample only. (In fact, the normal theory MLE and MVUE were also considered, and sometimes performed marginally better than BIE when the sampling distribution was not normal, but the differences were always very small.)
- (ii) The parametric empirical Bayes (PEB) estimator for normally distributed data, as derived by Kuczera (1982): this makes maximum use of the information in the full data set when the data are normally distributed.
- (iii) An extreme value theory estimator (EVTE), as discussed by Weissman (1978) and Boos (1984): the estimator is based on the k largest order statistics, and is valid for any distribution within the domain of attraction of the extreme value distribution with n.d.f.

given by $\exp[-e^{-x}]$ (which includes most common distributional forms such as normal, log-normal, gamma and Weibull). EVTE is thus a non-parametric sampling theory estimator. Boos discusses choice of k at length. Following his suggestion, we investigated the performance of EVTE using both $k = \max(m/5; 4)$ and $k = \max(4m\beta; 4)$. In fact there turned out to be very little difference in the performance of these two, and in the numerical results to follow we shall list simply the results for the better of the two.

The simulation studies for normally distributed samples were aimed at identifying any losses experienced in using NPEB as against parametric procedures based on known distributional properties. The studies with the contaminated normal distribution on the other hand, were to identify the gains achievable in using NPEB rather than the other procedures, when the distributional assumptions of the parametric estimators are not satisfied.

Each repetition of the Monte Carlo procedure involved the following steps:

(a) n pairs (μ_i, σ_i^2) , $i=1, \dots, n$ are generated randomly from the normal/inverse gamma distribution defined in DeGroot (1970, pp. 168-170), i.e.

(i) the random variable Z is generated from the gamma distribution with p.d.f.

$$f(z) = \frac{b^a z^{a-1} e^{-bz}}{\Gamma(a)} \quad 0 < z < \infty$$

where $a, b > 0$ are given constants;

(ii) σ_i^2 is defined as $1/z_i$

- (iii) μ_i is generated from the normal distribution with mean zero (there is no loss of generality here) and variance σ_i^2/τ , for given $\tau > 0$.

It is useful to note here briefly, a physical interpretation for the parameters a , b and τ . DeGroot shows that the above joint distribution on (μ, σ) is a conjugate prior for sampling from the normal distribution. It is thus possible to view τ as the equivalent size of sample from $N(\mu, \sigma^2)$, such that the prior and sample information have equal weight in estimating μ . Similarly, $2a$ is the corresponding equivalent sample size to give equal weight in estimating σ^2 . The parameter b is simply a scaling factor to ensure a desired expectation for σ , from the relationship $E[\sigma^2] = b/(a-1)$.

- (b) For each $i(=1, \dots, n)$ and $j(=1, \dots, m)$, a value for x_{ij} is generated from $N(\mu_i, \sigma_i^2)$ with probability $1 - \gamma$, or from $N(\mu_i, \lambda\sigma_i^2)$ with probability γ , where $\lambda > 1$ is a variance inflation factor, and γ a probability of contamination.
- (c) The true 2%, 5% and 10% quantiles for each population are computed: these are all of the form $\mu_i + \nu\sigma_i^2$, where for the combinations of γ and λ used in the studies (see below), ν takes on the following values:

	$\gamma=0$	$\gamma=0,2$ $\lambda=25$	$\gamma=0,1$ $\lambda=100$	$\gamma=0,2$ $\lambda=100$
10% :	1,2817	1,8081	1,5355	2,0479
5% :	1,6452	3,3941	2,3193	6,7449
2% :	2,0542	6,4078	8,4162	12,8155

- (d) The estimates according to each of the three estimators are computed from the x_{ij} , and the squared difference between the estimates and the true values obtained in (c) are computed for each population, for the 2%, 5% and 10% quantiles respectively.

The MSE values reported below were obtained by averaging over all populations and over all repetitions of the Monte Carlo procedure. Each estimate of the MSE is based on approximately 20 000 values, which appeared to be sufficient to ensure that the standard deviation of each estimate was around the 1%-2% of the value or less (cf. the results given in tables 1-3).

As a base case for comparison, the following set of parameter values were used:

Number of populations (n)	:	6
Sample size (m)	:	20
Prior distribution parameters	:	$a = 5$
		$b = 4$
		$\tau = 10$

From the comment after step (a) in the description of the Monte Carlo procedure, it follows that the true value of prior information relative to sample information is about one-half, for the uncontaminated normal case (since $2a = \tau = 10$). This implies that $\alpha_0 \approx m/2$; to simulate a situation of greater prior ignorance, the value chosen for ϵ was $1/3$, corresponding to a prior bound of $\alpha_0 \leq 3m$.

Note that the sample size of 20 is small relative to the quantiles being estimated.

The results for the four sets of (γ, λ) -values given earlier, including the estimated standard deviations of the estimates of the MSE, are displayed in

tables 1 (10% quantile), 2 (5%) and 3 (2%). The "non-parametric EB: naive estimate" is that which simply uses the corresponding quantiles of the posterior expectation of $F(x)$ directly.

The main observations which can be made from tables 1, 2 and 3 are quite self-evident. Even when the distributional assumptions of the parametric procedures (PEB and BIE) are fully satisfied, there is still relatively little lost in using the NPEB estimator for the 5% and 10% quantiles. Even for the 2% quantile (and recall that this requires the estimation of a 1 in 50 event from only 20 observations), the MSE of the two parametric procedures is between 65% and 70% of that for NPEB. On the other hand, when the distributional assumptions are not satisfied, NPEB is very nearly uniformly best, sometimes quite dramatically so (e.g. for the 10% quantile when $\lambda = 100$). As may have been expected, the extreme value theory estimator does not perform very well here, when the number of samples is small relative to the quantiles being estimated.

The conclusion is that the NPEB procedure gives considerable robustness in estimating quantiles from small amounts of data, without undue loss in efficiency if the parametric assumptions turn out to be satisfied. Furthermore, the NPEB estimator has a distinct edge over the naive non-parametric empirical Bayes method, which is justification for the somewhat greater degree of computation needed.

Further simulation studies were carried out to investigate, for three estimators (NPEB, PEB and BIE), the effect on MSE of changes in (i) the prior distribution of (μ_i, σ_i^2) , which represents degree of similarity between populations; (ii) the number of populations; and (iii) the sample size. The results are tabulated in tables 4, 5 and 6 for parameter settings as follows (all values not stated are as in the base case):

Table 1. Mean squared errors of estimation of 10% quantiles. (Base case. *)
 [Figures in brackets denote standard deviation of Monte Carlo estimate.]

Degree of contamination of normal distribution		ESTIMATOR				
Proportion contamination (γ)	Variance ratio (λ)	Non-parametric empirical Bayes		Parametric EB (PEB)	Best invariant (BIE)	Extreme value theory (EVTE)
		Basic form (NPEB)	Naive estimate			
0	1	0,091 (0,001)	0,110 (0,001)	0,088 (0,001)	0,094 (0,001)	0,109 (0,001)
0,2	25	0,648 (0,013)	0,776 (0,023)	1,972 (0,021)	2,471 (0,030)	1,384 (0,022)
0,1	100	0,718 (0,026)	0,699 (0,045)	7,541 (0,088)	9,244 (0,119)	3,590 (0,065)
0,2	100	3,21 (0,07)	3,93 (0,11)	14,15 (0,13)	16,70 (0,17)	7,85 (0,12)

* See text for details.

Table 2. Mean squared errors of estimation of 5% quantiles. (Base case.)*
[Figures in brackets denote standard deviation of Monte Carlo estimate.]

Degree of contamination of normal distribution		ESTIMATOR				
Proportion contamination (γ)	Variance ratio (λ)	Non-parametric empirical Bayes		Parametric EB (PEB)	Best invariant (BIE)	Extreme value theory (EVTE)
		Basic form (NPEB)	Naive estimate			
0	1	0,131 (0,002)	0,172 (0,002)	0,113 (0,001)	0,122 (0,001)	0,164 (0,002)
0,2	25	1,85 (0,02)	3,12 (0,05)	1,60 (0,02)	2,14 (0,03)	3,24 (0,04)
0,1	100	5,73 (0,10)	8,85 (0,21)	10,47 (0,12)	13,12 (0,17)	11,43 (0,20)
0,2	100	8,61 (0,10)	14,27 (0,19)	7,50 (0,07)	9,47 (0,11)	16,05 (0,17)

* See text for details.

Table 3. Mean squared errors of estimation of 2% quantiles. (Base case. *)
 [Figures in brackets denote standard deviation of Monte Carlo estimate.]

Degree of contamination of normal distribution		ESTIMATOR				
Proportion contamination (γ)	Variance ratio (λ)	Non-parametric empirical Bayes		Parametric EB (PEB)	Best invariant (BIE)	Extreme value theory (EVTE)
		Basic form (NPEB)	Naive estimate			
0	1	0,227 (0,003)	0,304 (0,004)	0,147 (0,002)	0,162 (0,002)	0,288 (0,003)
0,2	25	4,96 (0,05)	7,63 (0,08)	6,24 (0,05)	6,34 (0,05)	10,39 (0,08)
0,1	100	16,24 (0,16)	33,35 (0,40)	17,13 (0,13)	18,28 (0,14)	31,56 (0,28)
0,2	100	20,88 (0,23)	30,48 (0,35)	32,62 (0,27)	32,73 (0,27)	52,38 (0,42)

* See text for details.

Table 4. Effect on MSE for each estimator of the parameters of the prior distribution for (μ, σ^2) - expressed as an equivalent sample size for the uncontaminated normal distribution.

Quantile	Sampling distribution	Equivalent sample size of prior	ESTIMATOR		
			Non-parametric EB (NPEB)	Parametric EB (PEB)	Best invariant estimator (BIE)*
10%	Uncontaminated Normal	5	0,113	0,100	0,095
		10	0,091	0,088	0,094
		20	0,070	0,082	0,094
	10% contaminated Normal	5	0,98	7,42	9,36
		10	0,72	7,54	9,24
		20	0,59	7,63	9,22
5%	Uncontaminated Normal	5	0,169	0,131	0,124
		10	0,131	0,113	0,122
		20	0,100	0,103	0,122
	10% contaminated Normal	5	6,27	10,29	13,29
		10	5,73	10,47	13,12
		20	5,48	10,60	13,10
2%	Uncontaminated Normal	5	0,330	0,175	0,164
		10	0,227	0,147	0,162
		20	0,166	0,132	0,162
	10% contaminated Normal	5	19,73	17,74	18,48
		10	16,24	17,13	18,28
		20	14,81	16,92	18,19

* The BIE does not depend on the value of prior information. The estimates obtained in each separate Monte Carlo study are, however shown for comparison.

Table 5. Effect on MSE for each estimator of the number of populations.

Quantile	Sampling distribution	Number of populations	ESTIMATOR		
			Non-parametric EB (NPEB)	Parametric EB (PEB)	Best invariant estimator (BIE)*
10%	Uncontaminated Normal	4	0,096	0,089	0,094
		6	0,091	0,088	0,094
		10	0,084	0,088	0,094
	10% contaminated Normal	4	0,97	8,05	9,41
		6	0,72	7,54	9,24
		10	0,52	7,25	9,21
5%	Uncontaminated Normal	4	0,141	0,113	0,122
		6	0,131	0,113	0,122
		10	0,121	0,113	0,122
	10% contaminated Normal	4	7,20	11,26	13,37
		6	5,73	10,47	13,12
		10	4,54	10,01	13,06
2%	Uncontaminated Normal	4	0,251	0,148	0,162
		6	0,227	0,147	0,162
		10	0,204	0,148	0,161
	10% contaminated Normal	4	20,18	17,68	18,61
		6	16,24	17,13	18,28
		10	13,18	17,06	18,42

* The BIE does not depend on the number of populations. The estimates obtained in each separate Monte Carlo study are, however, shown for comparison.

Table 6. Effect on MSE for each estimator of the sample size per population.

Quantile	Sampling distribution	Sample size	ESTIMATOR		
			Non-parametric EB (NPEB)	Parametric EB (PEB)	Best invariant estimator (BIE)
10%	Uncontaminated Normal	20	0,091	0,088	0,094
		50	0,045	0,036	0,036
	10% contaminated Normal	20	0,72	7,54	9,24
		50	0,21	7,49	8,21
5%	Uncontaminated Normal	20	0,131	0,113	0,122
		50	0,064	0,047	0,047
	10% contaminated Normal	20	5,73	10,47	13,12
		50	2,72	10,22	11,33
2%	Uncontaminated Normal	20	0,227	0,147	0,162
		50	0,111	0,062	0,063
	10% contaminated Normal	20	16,24	17,13	18,28
		50	10,26	9,39	9,58

Table 4: The parameters a , b and τ take on the following sets of values: $(2, 5, 1, 5, 5)$, $(5, 4, 10)$ and $(10, 9, 20)$; this gives a weight of prior information for the normal case as equivalent to sample sizes of 5, 10 and 20 respectively. Note that in all cases

$$E[\sigma_i^2] = E[z^{-1}] = b/(a-1) = 1.$$

Table 5: The parameter n takes on values of 4, 6 and 10 respectively.

Table 6: The parameter m takes on values of 20 and 50 respectively.

These studies were carried out for the uncontaminated normal and the 10% contaminated normal ($\gamma=0,1$; $\lambda=100$) cases only.

Generally speaking, the previous conclusions carry through to all these cases. Both empirical Bayes procedures do show some gains (as is expected) as the populations become more similar (increasing equivalent sample value of prior), and as the number of populations increase. The gains for NPEB are, however, rather more substantial than for PEB. In fact the gains in using the correct parametric procedures for the 2% quantile in the uncontaminated normal case as given in table 3, are very nearly cancelled by having an equivalent sample size of prior of 20 (table 4) or having 10 populations (table 5). Increasing sample sizes benefit all methods; NPEB benefits most from this cause in the case of the 5% and 10% quantiles, but the parametric procedures gain slightly more here for the 2% quantile.

5. Comments on a real-world application

In order to evaluate the performance of the proposed method with real data, records of annual inflows into dams in the Western Transvaal were taken from the South African streamflow records compiled by Zucchini and Adamson (1984). These authors have shown that the log-normal distribution provides a consistently better fit to annual inflows into these

dams than any other of the standard distributions, and they have tabulated estimated parameters for this distribution, for each dam, using the full set of data available.

For our purposes, the Hartebeespoort Dam was excluded from consideration, as these inflows were an order of magnitude larger than for any other dam (and thus hardly "similar"). Data were used from all other dams (six in all, as listed in table 7) for which records over the same forty-year interval were available (July 1936 to June 1976). The data were divided into two sets, a training set of the first 20 years' data, and a test set. The estimated 10% and 5% quantiles were computed using NPEB and the log-normal distribution fitted to the training set. These were compared with

- (i) the corresponding sample quantiles in the test set;
- (ii) the corresponding sample quantiles in the full data set (i.e. all forty years); and
- (iii) the corresponding quantiles using the log-normal distributions given by Zucchini and Adamson.

These results are summarized in table 7. Quite evidently, there is little difference between the two sets of predictions. It also appears that the quantiles from a single 20 year period are very variable (in comparison to those from the 40 year period for example); for this reason it seems most appropriate to base comparisons on the quantiles from the full forty years. The total sums of squared differences between the predicted values and the 40 year quantiles are given in table 8, which includes a similar comparison between the predicted values and the complete log-normal fits of Zucchini and Adamson. Once again these values re-inforce the conclusion above, that the two predictors have approximately equivalent performance. (By way of contrast, the sums of squared differences between quantiles in the two 20-year sets are 14149 for the 10% quantile and 34765 for the 5% quantile respectively.)

Table 7. Comparison of predicted and actual quantiles of annual inflows (10^6 m^3) of six dams in the Western Transvaal (1936-1976).

Dam	10% Quantiles					5% Quantiles				
	Predicted		Actual			Predicted		Actual		
	NPEB	Log-normal	Test set only	Test and training sets	Log-normal from full set	NPEB	Log-normal	Test set only	Test and training sets	Log-normal from full set
Olifantsnek	27,9	23,2	62,9	28,5	30,4	40,9	29,5	111,9	62,9	43,0
Vietvlei	23,3	19,3	48,7	34,7	30,5	32,7	24,4	82,9	48,7	42,5
Buffelspoort	29,9	27,3	25,8	25,8	27,2	39,9	32,1	27,6	27,6	34,5
Bospoort	45,7	51,3	142,0	52,1	77,5	63,0	74,8	254,6	142,0	143,6
Marico-Bosveld	59,9	63,5	52,8	52,8	64,6	87,9	83,7	95,7	95,7	85,1
klein Marico	30,8	23,8	24,6	24,6	30,4	83,7	36,3	32,9	40,1	50,1

Closer examination of the data suggests that one of the potential advantages of the NPEB approach is not realized in this case, because of the non-independence of the data (high or low inflows tend to occur in the same year for all dams). This, of course, limits the ability of the empirical Dirichlet prior to obtain a good estimate of the tails of the predictive distribution. In order to examine the effect of this, the 40 years data were randomly shuffled, independently for each dam, and the estimations and comparisons repeated. (The effect of this shuffling is to randomize which data are allocated to the training and test sets respectively.) The sums of squared differences between actual and predicted values for this case are displayed in table 9. There does appear to be a distinct shift in favour of the NPEB procedure.

The purpose of this discussion has *not* been to suggest that in the present example, the NPEB is significantly better than the assumption of a log-normal distribution. In fact, given the strong evidence in favour of the data being log-normal, and the numerical results of section 4, any difference would be surprising. What has, however, been demonstrated is that even in relatively unfavourable situations (the non-independence of the data), the NPEB procedure works at least as well as (what appears to be) the correct parametric model. This certainly helps to validate the procedure on real-world data, and allows it to be used in practice with the assurance that nothing is lost thereby, while at the same time providing protection against unexpected sampling distributions, especially when sample data are particularly sparse.

TABLE 8. Sums of squared differences between actual and predicted values in table 7.

Quantile	Predictor	Actual values used	
		40-year quantile	Full log-normal model
10%	NPEB Log-normal fit	277	1099
		383	908
5%	NPEB Log-normal fit	7193	6720
		6400	5441

TABLE 9. Sums of squared differences between actual and predicted values with random reshuffling of data for each dam.

Quantile	Predictor	Actual values used	
		40-year quantile	Full log-normal model
10%	NPEB Log-normal fit	312	1795
		453	1790
5%	NPEB Log-normal fit	5029	4815
		6644	6447

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